CS7641 Unsupervised Learning — Reproducibility Sheet

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1. Overleaf project (read-only)

- **Overleaf (read-only):** https://www.overleaf.com/project/6a123a01f7534325be8dfacf
- **GitHub (GT Enterprise):** https://github.gatech.edu/tleung37/CS7641_ml_report3
- **Commit hash:** e7e515c2e87e67d2c48a623ea1658c38d25379c8
- **Seed:** 7641 (course number; propagated everywhere)

2. Dataset continuity (UL uses same Covertype sample as SL and OL)

The working dataset is **identical** to the SL and OL Reports:

- 20,000-instance stratified sample (`StratifiedShuffleSplit`, seed 7641) from UCI Forest Cover Type.
- Same 60/20/20 train/val/test split (12,000/4,000/4,000 instances).
- Preprocessing: `StandardScaler` on continuous columns 0–9 (elevation, aspect, slope, hydrology, hillshade, fire distance) fit on **training split only** for downstream NN evaluation. For exploratory clustering/DR: fit on full 20k sample. Binary columns 10–53 (wilderness, soil) left as 0/1.
- `covtype_25k.csv.gz` in `data/` is the same file generated by `report_1_sl/src/data_loader.py` and reused in `report_2_ol/` and `report_3_ul/`.

3. Exact run instructions (standard Linux machine)

```
# Python environment
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
# Key versions: torch==2.3.1  scikit-learn==1.5.0
#   numpy==1.26.4  pandas==2.2.2  matplotlib==3.9.0
#   seaborn==0.13.2  scipy==1.13.0  pyyaml==6.0
#   umap-learn==0.5.12

# R environment (for rsanity/ audit)
Rscript -e 'install.packages(c("bookdown","yaml","jsonlite","dplyr","
   ggplot2","knitr","kableExtra","readr","tibble","tidyr","scales","
   cluster"))'
```

3.2 Commands to reproduce every figure and table

```
# (a) Run the full pipeline end-to-end.
# Reads configs/config.yaml, writes results/{logs,tables,figures}/
python run_pipeline.py

# (b) Compile the UL report PDF (twice for cross-references).
cd report
pdflatex -interaction=nonstopmode UL_Report.tex
pdflatex -interaction=nonstopmode UL_Report.tex
cd ..

# (c) Rerun the R sanity check (independent audit of Python results).
Rscript -e 'rmarkdown::render("rsanity/sanity_check_UL.Rmd",␣output_dir="
   rsanity")'

# (d) Compile this reproducibility sheet.
cd repro
pdflatex -interaction=nonstopmode REPRO_UL.tex
```

3.3 Data Paths

`sklearn.datasets.fetch_covtype()` is called by `src/data_loader.py` on first run if `data/covtype_25k.csv.gz` is absent; the dataset (~75 MB) caches to `.cache/covtype/`. No manual download is required.

3.4 Random Seeds

Primary seed: `seed = 7641` (course number; same as SL/OL Reports), set in `configs/config.yaml` and propagated to `numpy.random.seed`, `random.seed`, `torch.manual_seed`, and CuDNN determinism flags. RP seed sweep: `[7641, 42, 123, 7, 2024]`. ICA seed sensitivity sweep: `[7641, 42, 123, 7, 2024]`.

3.5 Output Landing Locations

- `results/logs/eda_summary.json` — dataset dimensions and class counts.
- `results/logs/kmeans_sweep.json` — K-Means sweep metrics.
- `results/logs/kmeans_final.json` — K-Means final (k=7) metrics + contingency.
- `results/logs/gmm_sweep.json` — GMM sweep (AIC, BIC, ARI).
- `results/logs/gmm_final.json` — GMM final (n=7) metrics + contingency.
- `results/logs/pca_diagnostics.json` — EVR, cumulative variance, RMSE.
- `results/logs/ica_diagnostics.json` — kurtosis, seed sensitivity.
- `results/logs/rp_diagnostics.json` — distance-preservation across 5 seeds.
- `results/logs/umap_diagnostics.json` — UMAP hyperparameters + wall time.
- `results/logs/kmeans_after_{pca,ica,rp}.json` — Part 3 KMeans.
- `results/logs/gmm_after_{pca,ica,rp}.json` — Part 3 GMM.
- `results/logs/nn_after_dr.json` — Part 4 NN metrics for all 4 spaces.
- `results/tables/clustering_summary.csv` — compact clustering comparison.
- `results/tables/nn_dr_summary.csv` — compact NN comparison.
- `results/figures/*.pdf` — all 10 figures (white bg, denim palette).
- `rsanity/sanity_check_UL.pdf` — R audit output.

4. Clustering disclosures (Part 1)

Parameter	Value
K-Means k range	2, 3, 4, 5, 6, 7, 8, 10, 12
K-Means final k	7 (matches label count)
K-Means n_init	20
K-Means max_iter	500
GMM n range	2, 3, 4, 5, 6, 7, 8, 10, 12
GMM final n	7
GMM covariance	full
GMM n_init	5
GMM max_iter	200
Silhouette sample	3,000 (random subsample for speed)
Seed	7641

5. DR disclosures (Part 2)

Method	Parameter	Value
PCA	n_components (full)	54 (all)
PCA	n_components (final)	30 (99.5% cumvar)
ICA	n_components	20
ICA	max_iter / tol	500 / 1e-4
ICA	whiten	unit-variance
ICA	seed sweep	[7641, 42, 123, 7, 2024]
RP	n_components	30
RP	type	GaussianRandomProjection
RP	seed sweep	[7641, 42, 123, 7, 2024]
RP	dist-preservation sample	500 pairs
UMAP (extra)	n_components	2 (visualization only)
UMAP (extra)	n_neighbors	30
UMAP (extra)	min_dist	0.1

UMAP was used for visualization only. It was **not** used for downstream NN features because UMAP does not support standard transductive train/test transform without fitting on the test set.

6. NN-after-DR disclosures (Part 4)

Best justified recipe from SL and OL Reports:

Parameter	Value
Architecture	[256, 128] hidden, ReLU, BatchNorm
Optimizer	Adam no-bias-correction (OL Part2 best)
Learning rate	10^{-3}
Weight decay (L2)	10^{-4} (OL Part3 best single regularizer)
Batch size	256
Epochs	40 (max)
Early stopping	patience = 10, monitor val loss
Seed	7641
DR leakage control	fit on train split only, apply to val/test

Input dimension adjusted per DR method: original 54, PCA 30, ICA 20, RP 30. All other parameters held constant for fair comparison.

Adam without bias-correction update rule:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1-\beta_2)(g_t)^2, \\ \theta_t &= \theta_{t-1} - \alpha\, m_t/(\sqrt{v_t}+\varepsilon). \end{aligned}$$

Custom implementation in `src/nn_after_dr.py::AdamNoBias`.

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7. R sanity check — method and interpretation

The `rsanity/sanity_check_UL.Rmd` performs three audits:

1. **K-Means agreement.** Runs R `stats::kmeans` (k=7, nstart=10) on the same scaled 20k sample and compares silhouette to Python sklearn. Observed: R = 0.1322 vs. Python = 0.1306 ($|\Delta|$=0.0016 < 0.02 — **confirmed**).
2. **PCA agreement.** R `prcomp` vs. Python PCA: $n_{90\%}$=10, $n_{95\%}$=14, PC1-EVR = 0.226 — exact match.
3. **Log audit.** Loads all 10 clustering JSON logs (`kmeans/gmm_{final,after_pca,after_ica,after_rp}.json`) and verifies silhouette $\in [-1,1]$, ARI $\in [-1,1]$, NMI $\in [0,1]$. **All 10 logs pass.**

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8. Extended Clustering Table

Space	Algo	Sil	DB	CH	ARI	NMI
Original	K-Means	0.1306	1.792	2683	0.023	0.067
Original	GMM	−0.007	4.087	513	0.052	0.131
PCA-30	K-Means	0.1326	1.782	—	0.022	0.066
PCA-30	GMM	−0.001	3.501	—	0.117	0.186
ICA-20	K-Means	0.1416	1.701	—	−0.005	0.105
ICA-20	GMM	0.1304	1.780	—	0.027	0.143
RP-30	K-Means	0.1478	1.650	—	0.027	0.080
RP-30	GMM	0.013	3.402	—	0.125	0.170

9. Extended NN After DR Table

Features	dim	Macro-F1	Accuracy	Bal. Acc.	Train (s)
Original	54	0.6803	0.7940	0.6508	5.6
PCA-30	30	0.6951	0.8000	0.6642	7.0
ICA-20	20	0.6785	0.7875	0.6359	7.2
RP-30	30	0.6670	0.7883	0.6339	6.9

10. Hardware and software versions

- OS: Ubuntu / Debian Linux (x86_64)
- CPU: 2-core x86_64; no GPU used for submission numbers
- Python: 3.12.x
- Key packages: `torch 2.3.1`, `scikit-learn 1.5.0`, `numpy 1.26.4`, `pandas 2.2.2`, `matplotlib 3.9.0`, `seaborn 0.13.2`, `scipy 1.13.0`, `umap-learn 0.5.12`
- R: 4.5.0; `cluster 2.1`, `bookdown 0.40`, `jsonlite 1.8`, `ggplot2 3.5`, `kableExtra 1.4`
- LaTeX: TeX Live 2025 / Debian with IEEEtran conference class